LIBRARY MANAGEMENT SYSTEM

index.html

```
<!DOCTYPE html>

<html lang="en" ng-app="lmsApp">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Library Management System</title>

  <meta name="description" content="A comprehensive AngularJS library management system
for books, members, transactions, and fines." />

  <!-- Bootstrap 5 CSS -->

  <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" />

  <!-- Bootstrap Icons -->

  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.11.3/font/bootstrap-icons.min.css" />

  <!-- Google Fonts -->

  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=sw
ap" rel="stylesheet" />

  <!-- App Styles -->

  <link rel="stylesheet" href="css/app.css" />

</head>

<body>

  <!-- AngularJS ng-view renders routed views here -->

  <div ng-view></div>

  <!-- AngularJS 1.8 + ngRoute -->

  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
```

```
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular-
route.min.js"></script>
  <!-- Bootstrap 5 JS -->
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
  <!-- Core App Module + Routing -->
  <script src="app.js"></script>
  <!-- Services -->
  <script src="services/storage.service.js"></script>
  <script src="services/auth.service.js"></script>
  <script src="services/book.service.js"></script>
  <script src="services/member.service.js"></script>
  <script src="services/transaction.service.js"></script>
  <!-- Controllers -->
  <script src="controllers/login.controller.js"></script>
  <script src="controllers/dashboard.controller.js"></script>
  <script src="controllers/books.controller.js"></script>
  <script src="controllers/members.controller.js"></script>
  <script src="controllers/transactions.controller.js"></script>
</body>
</html>
```

app.js

```
angular.module('lmsApp', ['ngRoute'])
  .config(['$routeProvider', '$locationProvider', function ($routeProvider, $locationProvider) {
    $locationProvider.hashPrefix(''); // Fixes default #!/ routing to just #/
    $routeProvider
      .when('/login', {
        templateUrl: 'views/login.html',
```

```
  controller: 'LoginController',
  controllerAs: 'vm'
})
.when('/dashboard', {
  templateUrl: 'views/dashboard.html',
  controller: 'DashboardController',
  controllerAs: 'vm',
  resolve: {
    auth: ['AuthService', '$location', function (AuthService, $location) {
      if (!AuthService.isAuthenticated()) {
        $location.path('/login');
      }
    }]
  }
})
.when('/books', {
  templateUrl: 'views/books.html',
  controller: 'BooksController',
  controllerAs: 'vm',
  resolve: {
    auth: ['AuthService', '$location', function (AuthService, $location) {
      if (!AuthService.isAuthenticated()) {
        $location.path('/login');
      }
    }]
  }
})
```

```
    .when('/members', {
      templateUrl: 'views/members.html',
      controller: 'MembersController',
      controllerAs: 'vm',
      resolve: {
        auth: ['AuthService', '$location', function (AuthService, $location) {
          if (!AuthService.isAuthenticated()) {
            $location.path('/login');
          }
        }]
      }
    })
    .when('/transactions', {
      templateUrl: 'views/transactions.html',
      controller: 'TransactionsController',
      controllerAs: 'vm',
      resolve: {
        auth: ['AuthService', '$location', function (AuthService, $location) {
          if (!AuthService.isAuthenticated()) {
            $location.path('/login');
          }
        }]
      }
    })
    .otherwise({ redirectTo: '/login' });
}])
// Global run block — redirect unauthenticated users away from protected routes
```

```
  .run(['$rootScope', '$location', 'AuthService', 'StorageService',
    function ($rootScope, $location, AuthService, StorageService) {
      // Initialize localStorage with sample data on first run
      StorageService.initializeStore();
      $rootScope.$on('$routeChangeStart', function (event, next) {
        var isProtected = next && next.resolve && next.resolve.auth;
        if (isProtected && !AuthService.isAuthenticated()) {
          $location.path('/login');
        }
      });
    }
  ]);
```

login.controller.js

```
angular.module('lmsApp')
  .controller('LoginController', ['$scope', '$location', 'AuthService',
    function ($scope, $location, AuthService) {
      // Redirect if already logged in
      if (AuthService.isAuthenticated()) {
        $location.path('/dashboard');
      }
      $scope.credentials = { email: '', password: '' };
      $scope.error       = '';
      $scope.loading     = false;
      $scope.login = function () {
        $scope.loading = true;
        $scope.error   = '';
```

```
      var success = AuthService.login($scope.credentials.email, $scope.credentials.password);
      if (success) {
        $location.path('/dashboard');
      } else {
        $scope.error   = 'Invalid email or password.';
        $scope.loading = false;
      }
    };
    $scope.fillDemo = function (role) {
      var demos = {
        admin:     { email: 'admin@library.com',     password: 'admin123'     },
        librarian: { email: 'librarian@library.com', password: 'librarian123' },
        member:    { email: 'member@library.com',    password: 'member123'    }
      };
      if (demos[role]) {
        $scope.credentials = angular.copy(demos[role]);
      }
    };
  }
]);
```

dashboard.controller.js

```
angular.module('lmsApp')
  .controller('DashboardController', ['$scope', '$location', 'AuthService', 'TransactionService',
'BookService', 'MemberService',
    function ($scope, $location, AuthService, TransactionService, BookService, MemberService)
{
      $scope.currentUser        = AuthService.getCurrentUser();
      $scope.showNotif          = false;
```

```
    $scope.isActive = function(path) { return $location.path() === path; };
    $scope.stats              = TransactionService.getDashboardStats();
    $scope.recentTransactions = TransactionService.getTransactions().slice(-5).reverse();
    $scope.books              = BookService.getBooks();
    $scope.members            = MemberService.getMembers();
    $scope.lowStockBooks = $scope.books
      .filter(function (b) { return b.availableCopies <= 1; })
      .slice(0, 5);
    $scope.membersWithFines = $scope.members
      .filter(function (m) { return (m.totalFines - m.paidFines) > 0; });
    $scope.getBookTitle = function (id) {
      var b = $scope.books.find(function (b) { return b.id === id; });
      return b ? b.title : 'Unknown';
    };
    $scope.getMemberName = function (id) {
      var m = $scope.members.find(function (m) { return m.id === id; });
      return m ? m.name : 'Unknown';
    };
    $scope.pendingFine = function (member) {
      return (member.totalFines - member.paidFines).toFixed(2);
    };
    $scope.navigate = function (path) { $location.path(path); };
    $scope.logout = function () {
      AuthService.logout();
      $location.path('/login');
    };}
]);
```

auth.service.js

```
angular.module('lmsApp')
  .service('AuthService', ['StorageService', '$rootScope', function (StorageService, $rootScope) {
    var self = this;
    // In-memory current user (mirrors React useState)
    var _currentUser = null;
    (function init() {
      _currentUser = StorageService.get(StorageService.KEYS.CURRENT_USER, null);
    })();
    self.login = function (email, password) {
      var users = StorageService.get(StorageService.KEYS.USERS, []);
      var matched = users.find(function (u) {
        return u.email === email && u.password === password;
      });
      if (matched) {
        _currentUser = matched;
        StorageService.set(StorageService.KEYS.CURRENT_USER, matched);
        $rootScope.$broadcast('auth:login', matched);
        return true;
      }
      return false;
    };
    self.logout = function () {
      _currentUser = null;
      StorageService.set(StorageService.KEYS.CURRENT_USER, null);
      $rootScope.$broadcast('auth:logout');
    };
```

```
    self.getCurrentUser = function () {
      return _currentUser;
    };
    self.isAuthenticated = function () {
      return _currentUser !== null;
    };
    self.hasRole = function (roles) {
      if (!_currentUser) return false;
      return roles.indexOf(_currentUser.role) !== -1;
    };
  }]);
```

storage.service.js

```
angular.module('lmsApp')
  .service('StorageService', function () {
    var STORAGE_VERSION = '3';
    var VERSION_KEY = 'lms_storage_version';
    var KEYS = {
      USERS:        'lms_users',
      BOOKS:        'lms_books',
      MEMBERS:      'lms_members',
      TRANSACTIONS: 'lms_transactions',
      FINES:        'lms_fines',
      CURRENT_USER: 'lms_current_user'
    };
    var sampleUsers = [
      { id: 'user-1', email: 'admin@library.com',     password: 'admin123',     name: 'System Admin',
role: 'admin',     createdAt: new Date().toISOString() },
```

```
    { id: 'user-2', email: 'librarian@library.com', password: 'librarian123', name: 'John Librarian', role: 'librarian', createdAt: new Date().toISOString() },
    { id: 'user-3', email: 'member@library.com',    password: 'member123',    name: 'Jane Member',    role: 'member',    createdAt: new Date().toISOString() }
  ];
  var sampleBooks = [
    { id: 'book-1', title: 'The Great Gatsby',         author: 'F. Scott Fitzgerald', isbn: '978-0743273565', category: 'Fiction',    publisher: 'Scribner',              publishYear: 1925, totalCopies: 5, availableCopies: 3, location: 'A-1-01', createdAt: new Date().toISOString() },
    { id: 'book-2', title: 'To Kill a Mockingbird',    author: 'Harper Lee',          isbn: '978-0446310789', category: 'Fiction',    publisher: 'Grand Central Publishing', publishYear: 1960, totalCopies: 4, availableCopies: 2, location: 'A-1-02', createdAt: new Date().toISOString() },
    { id: 'book-3', title: 'Clean Code',               author: 'Robert C. Martin',    isbn: '978-0132350884', category: 'Technology', publisher: 'Prentice Hall',          publishYear: 2008, totalCopies: 3, availableCopies: 1, location: 'B-2-01', createdAt: new Date().toISOString() },
    { id: 'book-4', title: 'Design Patterns',          author: 'Gang of Four',        isbn: '978-0201633610', category: 'Technology', publisher: 'Addison-Wesley',         publishYear: 1994, totalCopies: 2, availableCopies: 2, location: 'B-2-02', createdAt: new Date().toISOString() },
    { id: 'book-5', title: '1984',                     author: 'George Orwell',       isbn: '978-0451524935', category: 'Fiction',    publisher: 'Signet Classic',         publishYear: 1949, totalCopies: 6, availableCopies: 4, location: 'A-1-03', createdAt: new Date().toISOString() }
  ];
  var sampleMembers = [
    { id: 'member-1', name: 'Alice Johnson', email: 'alice@email.com', phone: '+1-555-0101', address: '123 Main St, City',    membershipType: 'premium',  status: 'active', joinDate: '2024-01-15', expiryDate: '2025-01-15', totalFines: 15.00, paidFines: 15.00 },
    { id: 'member-2', name: 'Bob Smith',     email: 'bob@email.com',   phone: '+1-555-0102', address: '456 Oak Ave, Town',    membershipType: 'standard', status: 'active', joinDate: '2024-03-20', expiryDate: '2025-03-20', totalFines:  5.00, paidFines:  0    },
    { id: 'member-3', name: 'Carol Davis',   email: 'carol@email.com', phone: '+1-555-0103', address: '789 Pine Rd, Village', membershipType: 'student',  status: 'active', joinDate: '2024-06-01', expiryDate: '2025-06-01', totalFines:  0,    paidFines:  0    }
  ];
```

```
var sampleTransactions = [
  { id: 'trans-1', bookId: 'book-1', memberId: 'member-1', issueDate: '2024-12-01', dueDate: '2024-12-15', returnDate: '2024-12-20', status: 'returned', fine: 5.00, finePaid: true  },
  { id: 'trans-2', bookId: 'book-2', memberId: 'member-2', issueDate: '2024-12-10', dueDate: '2024-12-24', returnDate: null,         status: 'overdue',  fine: 5.00, finePaid: false },
  { id: 'trans-3', bookId: 'book-3', memberId: 'member-1', issueDate: '2025-01-05', dueDate: '2025-01-19', returnDate: null,         status: 'issued',   fine: 0,    finePaid: false }
];
this.get = function (key, defaultValue) {
  var storedVersion = localStorage.getItem(VERSION_KEY);
  if (storedVersion !== STORAGE_VERSION) {
    // Version mismatch — wipe stale data
    Object.values(KEYS).forEach(function (k) { localStorage.removeItem(k); });
    localStorage.setItem(VERSION_KEY, STORAGE_VERSION);
    return defaultValue;
  }
  var raw = localStorage.getItem(key);
  if (raw) {
    try { return JSON.parse(raw); } catch (e) { return defaultValue; }
  }
  return defaultValue;
};
this.set = function (key, value) {
  if (!localStorage.getItem(VERSION_KEY)) {
    localStorage.setItem(VERSION_KEY, STORAGE_VERSION);
  }
  localStorage.setItem(key, JSON.stringify(value));
}
```

```
    this.initializeStore = function () {
      var self = this;
      if (!localStorage.getItem(KEYS.USERS))        self.set(KEYS.USERS,        sampleUsers);
      if (!localStorage.getItem(KEYS.BOOKS))        self.set(KEYS.BOOKS,        sampleBooks);
      if (!localStorage.getItem(KEYS.MEMBERS))      self.set(KEYS.MEMBERS,
sampleMembers);
      if (!localStorage.getItem(KEYS.TRANSACTIONS)) self.set(KEYS.TRANSACTIONS,
sampleTransactions);
      if (!localStorage.getItem(KEYS.FINES))        self.set(KEYS.FINES,        []);
    };
    this.generateId = function (prefix) {
      return (prefix || 'id') + '-' + Date.now() + '-' + Math.random().toString(36).substr(2, 9);
    };
  });
```